

|            |                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------|
| EXPT NO:4A | <b>EXPLORATORY DATA ANALYSIS (EDA) – DATA INSPECTION, FILTERING, STATISTICAL ANALYSIS AND CORRELATION</b> |
|            |                                                                                                           |

**Aim:**

To perform data inspection, filtering, grouping, aggregation, descriptive statistical analysis, and correlation analysis using Pandas and visualize the results using Matplotlib.

**Algorithm:**

1. Import the required Pandas, NumPy and Matplotlib libraries.
2. Read the stud\_data.csv file using Pandas.
3. Display the dataset and inspect the first few records.
4. Find the number of rows and columns and display the data types.
5. Generate the statistical summary of the dataset.
6. Filter the students based on marks and attendance conditions.
7. Sort the student records based on marks and attendance.
8. Group the data based on department and calculate average marks and attendance.
9. Calculate mean, median, minimum, maximum and standard deviation of marks.
10. Calculate the correlation between marks and attendance.
11. Create bar chart, line chart, scatter plot, histogram and box plot using Matplotlib.
12. Display the results and interpret the obtained data.

**Program:**

```
# Import libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

# Read dataset

df = pd.read_csv("stud_data.csv")

# Display dataset

print(df)

# Display first five records

print(df.head())

# Display shape

print(df.shape)

# Display data types

print(df.dtypes)

# Display statistical summary

print(df.describe())

# Filter students with marks greater than 80
```

```
print(df[df["Marks"] > 80])
```

```
# Filter students with attendance greater than 90
```

```
print(df[df["Attendance"] > 90])
```

```
# Sort students by marks
```

```
print(df.sort_values("Marks", ascending=False))
```

```
# Sort students by attendance
```

```
print(df.sort_values("Attendance", ascending=False))
```

```
# Department-wise average marks
```

```
print(df.groupby("Department")["Marks"].mean())
```

```
# Department-wise average attendance
```

```
print(df.groupby("Department")["Attendance"].mean())
```

```
# Calculate mean
```

```
print("Mean Marks:", df["Marks"].mean())
```

```
# Calculate median
```

```
print("Median Marks:", df["Marks"].median())
```

```
# Calculate minimum
```

```
print("Minimum Marks:", df["Marks"].min())
```

```
# Calculate maximum
print("Maximum Marks:", df["Marks"].max())

# Calculate standard deviation
print("Standard Deviation:", df["Marks"].std())

# Display correlation matrix
print(df[["Marks", "Attendance"]].corr())

# Display correlation between marks and attendance
print("Correlation:", df["Marks"].corr(df["Attendance"]))

# Bar chart
plt.bar(df["Name"], df["Marks"])
plt.title("Student Marks")
plt.xlabel("Name")
plt.ylabel("Marks")
plt.show()

# Line chart
plt.plot(df["Name"], df["Attendance"], marker="o")
plt.title("Student Attendance")
plt.xlabel("Name")
plt.ylabel("Attendance")
```

```
plt.show()
```

```
# Scatter plot
```

```
plt.scatter(df["Attendance"], df["Marks"])
```

```
plt.title("Marks vs Attendance")
```

```
plt.xlabel("Attendance")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

```
# Histogram
```

```
plt.hist(df["Marks"], bins=5)
```

```
plt.title("Marks Distribution")
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Number of Students")
```

```
plt.show()
```

```
# Box plot
```

```
plt.boxplot(df["Marks"])
```

```
plt.title("Marks Box Plot")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

Output:

|   | Roll No | Name    | Department | Marks | Attendance |
|---|---------|---------|------------|-------|------------|
| 0 | 101     | Arun    | CSE        | 85    | 92         |
| 1 | 102     | Bala    | IT         | 78    | 88         |
| 2 | 103     | Divya   | CSE        | 95    | 96         |
| 3 | 104     | Priya   | AI&DS      | 90    | 94         |
| 4 | 105     | Ravi    | IT         | 72    | 85         |
| 5 | 106     | Karthik | AI&DS      | 88    | 91         |

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(6, 5)

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

|       | Roll No    | Marks     | Attendance |
|-------|------------|-----------|------------|
| count | 6.000000   | 6.000000  | 6.00       |
| mean  | 103.500000 | 84.666667 | 91.00      |
| std   | 1.870829   | 8.382521  | 4.00       |
| min   | 101.000000 | 72.000000 | 85.00      |
| 25%   | 102.250000 | 79.750000 | 88.75      |
| 50%   | 103.500000 | 86.500000 | 91.50      |
| 75%   | 104.750000 | 89.500000 | 93.50      |
| max   | 106.000000 | 95.000000 | 96.00      |

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Department

AI&DS 89.0

CSE 90.0

IT 75.0

Name: Marks, dtype: float64

Department

AI&DS 92.5

CSE 94.0

IT 86.5

Name: Attendance, dtype: float64

Mean Marks: 84.66666666666667

Median Marks: 86.5

Minimum Marks: 72

Maximum Marks: 95

Standard Deviation: 8.382521498133285

Marks Attendance

Marks 1.000000 0.978226

Attendance 0.978226 1.000000

Correlation: 0.9782259433305447





